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Agentic AI Hackathon 2026

SAP + AWS AGENTIC AI HACKATHON 2026 · SUPPLY CHAIN

SIGAP

Supply disruptions handled before anyone has to notice

Twelve agents read your SAP data, obey Indonesia's local-content and import rules, and put one decision you can approve on your desk.

Team Socrates | APP Group | Live demo

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APP

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01 · THE PROBLEM

A port closes Monday. The decision lands Thursday.

1 **The news arrives on the wrong channel**
A headline, a WhatsApp group, a call from the forwarder.

2 **Three days of tracing it by hand**
Excel, email, asking around: which orders are actually hit?

3 **The decision is made on stale data**
The numbers already moved on Tuesday.

4 **Then the reasoning disappears**
Nothing records why. An audit becomes archaeology.

WHERE THE MONEY ACTUALLY GOES

The fastest decision is often the one that was never allowed.

This is not a missing-ERP problem.

The data is already in S/4HANA. What is missing are the hours between the signal and the decision.



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01 · THE PROBLEM

Two Indonesian rules that make the cheapest option illegal

TKDN — local content

Regulation and most contracts set a floor on how much must be made in Indonesia. Switching to an imported supplier pushes you under it.

LARTAS — import restrictions

A new country of origin needs permits, and permits do not arrive in a week.

Making the decision faster without enforcing these two is not progress. It is faster contract breach.

FROM THE LIVE DEMO

Option E — Rp 48 million, the cheapest — is struck out.

Local content falls to 34.8%.

Not because it costs too much. Because it is not allowed — and a price cannot overrule a rule.



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02 · THE SOLUTION

Twelve agents, one decision you can sign

01 **It watches instead of waiting**
Public hazard feeds, every few minutes, nobody asked.

02 **Twelve agents read the real data**
S/4HANA, master data, your uploads, the feeds.

03 **Kira strikes out what is not allowed**
Local content and import rules, before price gets a turn.

04 **Tara puts a number on each option**
Cost, stock-out date, arrival date, local-content impact.

05 **It acts inside its authority**
Above that limit it stops and asks a person.

ONE AGENT HOLDS A VETO

Kira blocks at two gates — when the option is chosen, and again when the PO is issued.

The veto is code reading a rule pack, not a prompt. A cheaper option cannot talk its way past it.



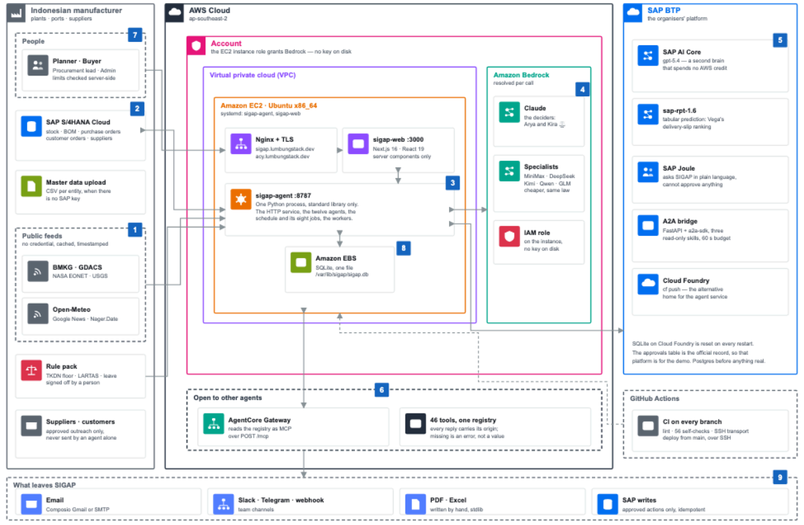
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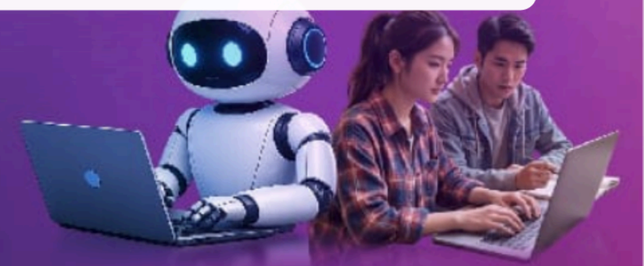
03 · ARCHITECTURE

AWS for the thinking, SAP for the data and the doors



Agent service: Python 3, standard library only — `http.server` + `sqlite3`. This diagram is generated by `sigap/architecture/diagram.py`, not drawn.

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PAUSE

Demo.

One typhoon closes Ningbo. Watch SIGAP find it, size it, strike out the cheapest option, and then stop to ask for a signature.

WHAT YOU ARE ABOUT TO SEE

Live, not a recording — against real SAP S/4HANA and Amazon Bedrock.

Every number on screen can be recomputed from the commands in DEMO.md.



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04 · DETECT

It is already watching, all night

01 **Five public feeds, no licence needed**
BMKG, GDACS, NASA EONET, USGS, Open-Meteo.

02 **Eight jobs on their own clock**
Quakes every five minutes; cyclones, floods and marine weather on theirs.

03 **It watches your paperwork too**
Local-content certificates that are about to expire, before they lapse.

04 **One digest at 07:30**
What moved overnight, and what is waiting for your signature.

NOTHING TO REMEMBER TO CHECK

No inbox to watch, no dashboard to open, for SIGAP to start working.

It only needs your ports, plants and material list: every event is matched against them by distance, so a quake 2,000 km away stays silent.



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05 · ACT

It does the work, not just the advice

Buys inside a limit

Under Rp 50 million it moves on its own. Above it, it stops and asks. One setting, checked on the server.

Writes back into SAP

The approved action leaves as a purchase order — idempotent, and only ever after a person signs off.

Talks to your suppliers

It drafts the outreach email and holds it. No agent sends anything on its own.

Tells the right people

Email, Slack, Telegram, or a webhook for Teams and ticket queues.

WHERE IT STOPS

Every action that leaves the building has a gate a human can hold shut.

Approve, hold or reject — and the record keeps who did it, when, and on which figures.



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06 · OPEN

Built for other agents, not only for people

MCP — All 46 tools over one POST /mcp endpoint. AWS AgentCore Gateway speaks MCP, and so does SAP Joule.

A2A — An agent card with three read-only skills: investigate a disruption, check how one is going, scan for risk. None of them can approve anything.

Reports — PDF and Excel per handling, plus a portfolio view across all of them — written with the standard library, no reporting dependency.

WHY THAT MATTERS HERE

SIGAP is a teammate other agents can call, not an app you have to open.

Ask Joule in plain language and the answer still carries where each figure came from — and Joule is not allowed to approve.



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ONE SENTENCE TO REMEMBER

Not an agent that decides faster — an agent that refuses to decide what it may not.

A decision you can approve, with the origin of every figure behind it, in minutes — and never against local-content or import rules.

`sigap.lumbungstack.dev`

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TEAM SOCRATES

Ready for your questions.

Thank you — and please ask us to re-run any command whose numbers you saw on screen.

